

## RSCTF SCORING DESIGN PAPER

# TWO WAYS TO HOLD THE HILL: BOOT2ROOT KOTH AND LEADERBOARD KOTH

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Format proposal · Version 1.0 · 30 July 2026

One fixed formula per format · no organizer-selected scoring versions

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**ABSTRACT**

*This paper names and separates two King of the Hill competition formats. **Boot2Root KoH** is an exclusive-control contest over one shared machine: teams acquire the current hill, retain it, and remain responsible for service health. **Leaderboard KoH** is a concurrent application or protocol contest: every eligible team can play in the same tick, RSCTF normalizes challenge-native activity and objective evidence, and repeated first-place results earn a bounded sustained-lead bonus. The name “Leaderboard” describes the competitive object without restricting the challenge to HTTP APIs. The proposed formula never penalizes failed exploitation attempts as a score component. Evidence authenticity remains an admission decision enforced by scoped capabilities, signed referee snapshots, replay fencing, exact runtime context, and an independent functional checker. Both formats produce a bounded 0–100 hill score before common hill and epoch normalization.*

**Keywords:** *King of the Hill; Boot2Root; Leaderboard KoH; normalized objectives; sustained lead; cybersecurity competition; RSCTF*

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**Status:** design proposal. RSCTF **v0.1.31** calls the concurrent format “API arena” and still applies an integrity ratio to its tick score. This paper specifies the proposed replacement name and no-integrity-points formula. It is not the live scoring contract until the backend, UI, tests, and organizer documentation adopt it together.

## 1. THE FORMAT DECISION

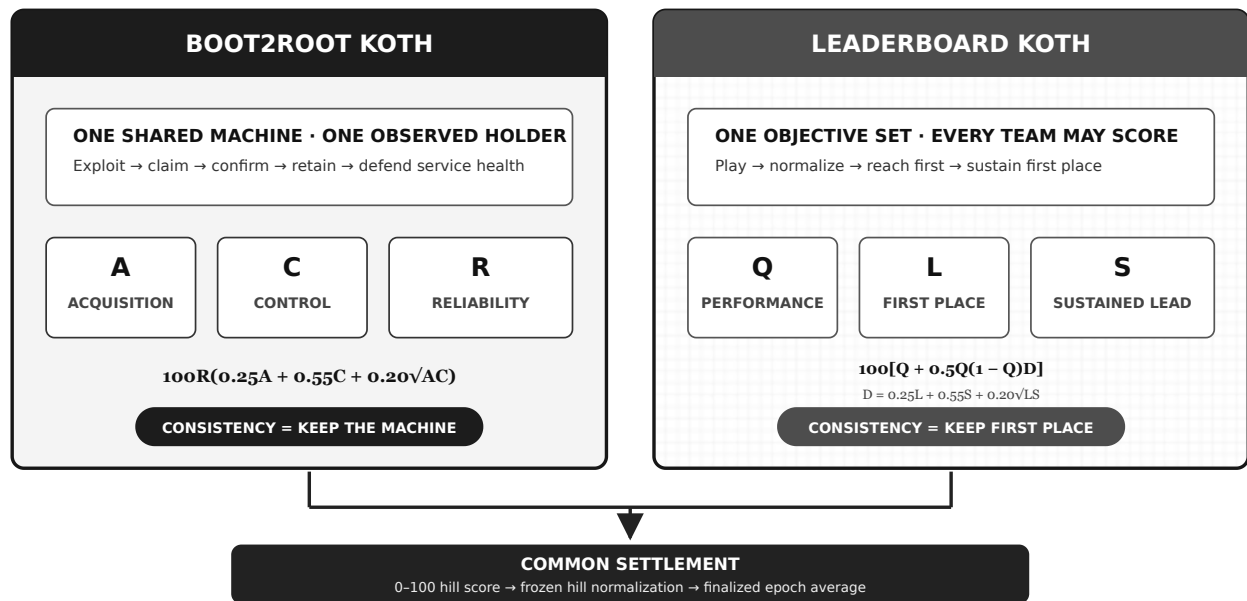
RSCTF should expose two KoH types because they answer different competitive questions:

- **Boot2Root KoH:** “Who controls this shared machine?”
- **Leaderboard KoH:** “Who performs best, and who can remain first?”

The first question has one observed holder. The second permits every team to score concurrently. A transport label such as “API” does not express that difference: a Leaderboard hill may be an HTTP service, binary protocol, simulator, game server, cryptographic service, or another instrumented application.

## Two meanings of “hold the hill”

Exclusive machine control and concurrent leaderboard dominance need different evidence



**Figure 1.** Boot2Root KoH measures exclusive machine control. Leaderboard KoH measures normalized concurrent performance and sustained first place. Both pass through the same bounded hill and epoch settlement.

**Table 1.** Format contract.

| Property              | Boot2Root KoH                       | Leaderboard KoH                                         |
|-----------------------|-------------------------------------|---------------------------------------------------------|
| Competitive object    | One shared machine                  | One application or protocol objective set               |
| Simultaneous scorers  | At most one observed controller     | Every eligible team                                     |
| Primary player action | Exploit, claim, and retain the host | Complete meaningful actions and optimize objectives     |
| Core evidence         | Acquisition, control, reliability   | Activity, objective performance, first-place continuity |
| Meaning of “king”     | Current confirmed controller        | Current normalized leaderboard leader                   |
| Failure experiments   | Expected hacking behavior           | Expected hacking behavior                               |
| Evidence fraud        | Rejected or adjudicated             | Rejected or adjudicated                                 |
| Local score           | <b>0–100</b>                        | <b>0–100</b>                                            |

## 2. WHY “LEADERBOARD KOTH”

The proposed name has three useful properties.

First, it names what teams are trying to hold: first place on a challenge-local leaderboard. Second, it remains accurate when the challenge is not an API. Third, it gives the UI unambiguous language. A **Boot2Root** badge means that a holder, provisional claim, confirmation, and takeover state exist. A **Leaderboard** badge means that several teams can score in one tick and that the board shows normalized performance plus sustained-lead evidence.

“Arena” remains a useful description, but it does not tell players what constitutes control. “Objective KoH” describes the inputs but not the first-place continuity requested by the format. “Leaderboard KoH” covers both.

### 3. BOOT2ROOT KOTH

#### 3.1 Competition protocol

One managed hill exposes a machine or service that teams attack. RSCTF issues each eligible team a high-entropy capability for the exact hill, crown cycle, reset attempt, and container. A team claims the hill by placing its current capability in the marker location after obtaining the required access.

The checker reads the marker before and after a functional service probe. A stable eligible capability establishes control evidence. Confirmation requires the configured healthy observation streak, so a transient write does not become a capture. The controlling team remains responsible for service health until another team takes over or the clean reset begins.

#### 3.2 Fixed score

For one team, hill, and epoch:

- $A$  is the confirmed acquisition rate;
- $C$  is the fraction of personally eligible ticks controlled; and
- $R$  is service reliability while the team is responsible.

The constant Boot2Root score is:

$$B^{\text{root}} = 0.25A + 0.55C + 0.20\sqrt{AC},$$

$$H^{\text{root}} = 100RB^{\text{root}}.$$

Control has the largest direct coefficient. The geometric term rewards teams that both acquire and retain the machine, while reliability constrains the complete result. The formula does not change between events.

#### 3.3 Meaning of consistent leadership

Consistency is physical and exclusive in Boot2Root KoH. A team earns it by keeping the current capability on the machine across randomized healthy checks. Another team cannot hold the same hill during those ticks.

Pristine crown-cycle replacement keeps that control contest repeatable. The old container is destroyed, old capabilities are revoked, and the same frozen image starts clean before the next cycle.

### 4. LEADERBOARD KOTH

#### 4.1 Competition protocol

Every eligible team uses the same player-facing application or protocol during the same scoring tick. The challenge records capability-bound events. A trusted referee outside the attacker-controlled runtime converts those events into bounded integer ratios and signs one complete field snapshot. It submits evidence, never points.

Players may probe, fuzz, crash their own session, send malformed inputs, and fail while developing an exploit unless the published event rules prohibit a specific action. Those attempts are part of hacking. RSCTF therefore does not use “clean play” or a valid-attempt percentage as a score multiplier.

#### 4.2 Native-score normalization

For team  $i$ , hill  $h$ , and tick  $t$ , let:

$$E_{iht} = \frac{\text{meaningful activity completed}}{\text{published activity target}},$$

and let objective  $j$  report bounded integer evidence  $o_{ihtj}^+ / o_{ihtj}^*$ . RSCTF independently normalizes each objective:

$$p_{ihtj} = \frac{o_{ihtj}^+}{o_{ihtj}^*}, \quad P_{iht} = \frac{1}{m} \sum_{j=1}^m p_{ihtj}, \quad 1 \leq m \leq 16.$$

A native result of **9/10** and another of **900/1000** both become **0.9**. Numerical scale cannot become accidental weight. The challenge must publish what earns each numerator and how each denominator is fixed.

Activity counts completed, capability-bound actions rather than request volume. Suitable units include distinct tasks completed, protocol phases reached, valid transactions committed, unique puzzles solved, or verified work units. Raw HTTP requests, packets, or connection counts are unsuitable because spam would become a scoring strategy.

#### 4.3 Per-tick performance

The fixed performance core is a weighted harmonic mean:

$$B_{iht} = \begin{cases} 0, & E_{iht} = 0 \text{ or } P_{iht} = 0, \\ \frac{1}{0.35 / E_{iht} + 0.65 / P_{iht}}, & \text{otherwise.} \end{cases}$$

Both participation and objective performance are necessary. Objective performance has greater influence, but a team cannot replace activity with one excellent output. RSCTF stores this result for each tick before aggregation, so activity from one moment cannot combine with performance from another.

For  $T$  field-scorable ticks in an epoch:

$$Q_{ihe} = \frac{1}{T} \sum_{t=1}^T B_{iht}.$$

$Q$  is the absolute challenge-performance rate. It does not depend on another team's native point scale.

#### 4.4 First-place and sustained-lead evidence

A competitive tick requires at least two teams with positive performance. Within such a tick, the highest stored  $B$  receives lead credit. If  $k$  teams share the exact highest result, each receives  $1 / k$  rather than an arbitrary team-ID tie-break.

Let  $\ell_{iht}$  be that lead credit. Lead coverage and sustained lead are:

$$L_{ihe} = \frac{1}{T} \sum_{t=1}^T \ell_{iht},$$

$$S_{ihe} = \begin{cases} 0, & T < 2, \\ \frac{1}{T-1} \sum_{t=2}^T \min(\ell_{ih,t-1}, \ell_{iht}), & T \geq 2. \end{cases}$$

$L$  rewards reaching first place.  $S$  rewards remaining first across adjacent ticks. Five scattered wins and five consecutive wins therefore produce different consistency evidence.

The dominance rate deliberately mirrors the shape of the Boot2Root acquisition/control core:

$$D_{ihe} = 0.25L_{ihe} + 0.55S_{ihe} + 0.20\sqrt{L_{ihe}S_{ihe}}.$$

#### 4.5 Bounded local score

The proposed Leaderboard hill score is:

$$H_{ihe}^{\text{lead}} = 100[Q_{ihe} + 0.5Q_{ihe}(1 - Q_{ihe})D_{ihe}].$$

The first term pays for absolute performance. The second is a bounded bonus for sustained first place. It cannot create points when  $Q = 0$ , cannot push a perfect result above 100, and never raises the score beyond the **[0,100]** range. Its largest possible contribution is 12.5 points, reached at  $Q = 0.5$  and  $D = 1$ .

This structure avoids two common errors. A fixed ten-point winner award could make weak performance valuable in an empty field. Multiplying by an uncapped rank bonus could collapse several strong teams at the 100-point ceiling. The bounded formula scales the bonus by both achieved performance and remaining headroom.

### 5. WORKED CONSISTENCY EXAMPLE

Assume four teams each finish an epoch with  $Q = 0.80$  but reach first place in different patterns over ten ticks.

**Table 2.** Sustained leadership changes the bonus, not the underlying performance.

| Pattern                          | $L$   | $S$   | $D$   | Final score |
|----------------------------------|-------|-------|-------|-------------|
| Never first                      | 0.000 | 0.000 | 0.000 | 80.00       |
| First on five alternating ticks  | 0.500 | 0.000 | 0.125 | 81.00       |
| First for five consecutive ticks | 0.500 | 0.444 | 0.464 | 83.71       |
| First for all ten ticks          | 1.000 | 1.000 | 1.000 | 88.00       |

The alternating team reaches first as often as the five-tick leader, but it never retains first place across adjacent ticks. The consecutive leader earns the larger bonus. A team that remains first for the whole epoch receives the maximum dominance rate.

The bonus does not replace challenge performance. At  $Q = 0.20$  and  $D = 1$ , the result is 28 rather than a winner jackpot:

$$100[0.20 + 0.5(0.20)(0.80)(1)] = 28.$$

## 6. HARD TO CHEAT, FREE TO HACK

The scoring boundary must separate exploratory behavior from forged evidence.

### 6.1 Hacking behavior

These actions do not directly reduce points:

- failed exploit attempts;
- fuzzing and malformed challenge inputs;
- restarting or corrupting a team-owned session;
- automation permitted by the event rules; and
- trying an objective without completing it.

An unsuccessful attempt simply produces no completed activity or objective credit. Organizers may still rate-limit traffic to protect shared infrastructure, but that is an operational control rather than a scoring metric.

### 6.2 Evidence admission

These conditions reject or void evidence instead of applying a percentage penalty:

- invalid HMAC or revoked referee credential;
- replayed body or non-monotonic timestamp;
- stale capability hash;
- wrong game, challenge, target, round, cycle, reset, or container;
- objective count that differs from the frozen challenge scheme;
- incomplete field snapshot;
- snapshot change during the independent functional probe; and
- platform-wide checker or readiness failure.

One omitted team inside an otherwise complete field snapshot receives an explicit zero. A missing or untrustworthy whole-field snapshot is void for everyone. Neither case carries an earlier score forward.

### 6.3 Rule violations

Collusion, attacking platform infrastructure, targeting organizer systems, or other prohibited conduct belongs in the event's incident and sanction process. A hidden fractional multiplier is too ambiguous for these decisions. The organizer should preserve evidence, notify affected teams, and apply the published warning, score correction, or disqualification procedure.

## 7. SHARED SETTLEMENT

Both formats produce one local hill score in **[0, 100]**. RSCTF then applies the same bounded hill and epoch aggregation.

Let  $w_h \in [0.8, 1.2]$  be the frozen hill weight and  $z_{he}$  indicate whether hill  $h$  has field-scorable evidence in epoch  $e$ :

$$E_{ie} = \frac{\sum_h z_{he} w_h H_{ihe}}{\sum_h z_{he} w_h}.$$

A wholly void hill contributes to neither numerator nor denominator. A complete evidence-bearing epoch has weight one. A shortened final epoch uses the fraction of configured ticks that were played:

$$T_i = \frac{\sum_e q_e E_{ie}}{\sum_e q_e}.$$

There is no late-event multiplier. **Projected** may include open evidence; **Settled** includes finalized epochs and determines official rank.

## 8. ORGANIZER SELECTION GUIDE

Choose **Boot2Root KotH** when all of the following are true:

- the intended skill is obtaining privileged access to one shared target;
- exactly one team should control the target at a time;
- takeover and persistence are central to the challenge;
- the service checker can attribute health responsibility to the holder; and
- clean resets preserve the intended exploit path.

Choose **Leaderboard KotH** when these conditions fit better:

- several teams should play and score simultaneously;
- the challenge exposes measurable application or protocol outcomes;
- at least one meaningful activity unit and one objective can be independently verified;
- first place and sustained first place should matter; and
- a trusted referee can operate outside the player-controlled runtime.

Do not choose Leaderboard KotH merely because a challenge has an HTTP API. Use it when concurrent normalized competition is the intended game.

## 9. PROPOSED MIGRATION FROM API ARENA

The name and formula should change atomically in a future implementation:

1. rename the player-facing format from **API arena** to **Leaderboard KotH**, migrate the stored source identifier, and remove the superseded naming and scoring branch;
2. remove integrity from scoring and keep authenticity as an evidence gate;
3. add immutable per-tick lead credit and adjacent-tick sustained-lead evidence;
4. compute dominance and the final score only from finalized, field-scorable ticks;
5. update board columns to **Activity**, **Objectives**, and **Sustained lead**;
6. preserve the existing Boot2Root formula without scoring versions; and
7. migrate or void incompatible open API-arena evidence rather than reinterpreting it under the new policy.

The upgrade must run while no affected event is scoring. Settled numeric rollups remain immutable historical results; RSCTF does not recompute them and does not retain the superseded formula as a selectable scoring version.

## 10. PLAYER SUMMARY

BOOT2ROOT KOTH

Exploit the machine → claim it → keep it healthy → retain control

LEADERBOARD KOTH

Complete real actions → optimize every objective → reach first → stay first

EVIDENCE SECURITY

Authentic current evidence is accepted; forged or stale evidence is rejected

The two formats share the KotH idea but not the meaning of control. Boot2Root control is possession of a shared machine. Leaderboard control is sustained first place in a normalized concurrent challenge.

## APPENDIX A. SYMBOLS

**Table 3.** Symbols used in the two-format scoring proposal.

| Symbol | Definition                                        |
|--------|---------------------------------------------------|
| $A$    | Boot2Root confirmed acquisition rate              |
| $C$    | Boot2Root controlled-tick rate                    |
| $R$    | Boot2Root service reliability while responsible   |
| $E$    | Leaderboard meaningful activity rate for one tick |
| $P$    | Equal-weight mean of normalized objective ratios  |
| $B$    | Leaderboard per-tick harmonic performance         |
| $Q$    | Mean per-tick performance in one hill-epoch       |
| $\ell$ | Exact first-place credit for one competitive tick |
| $L$    | First-place coverage                              |
| $S$    | Adjacent-tick sustained-lead rate                 |
| $D$    | Combined Leaderboard dominance rate               |
| $H$    | Local hill score in $[0, 100]$                    |

## REFERENCES

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